

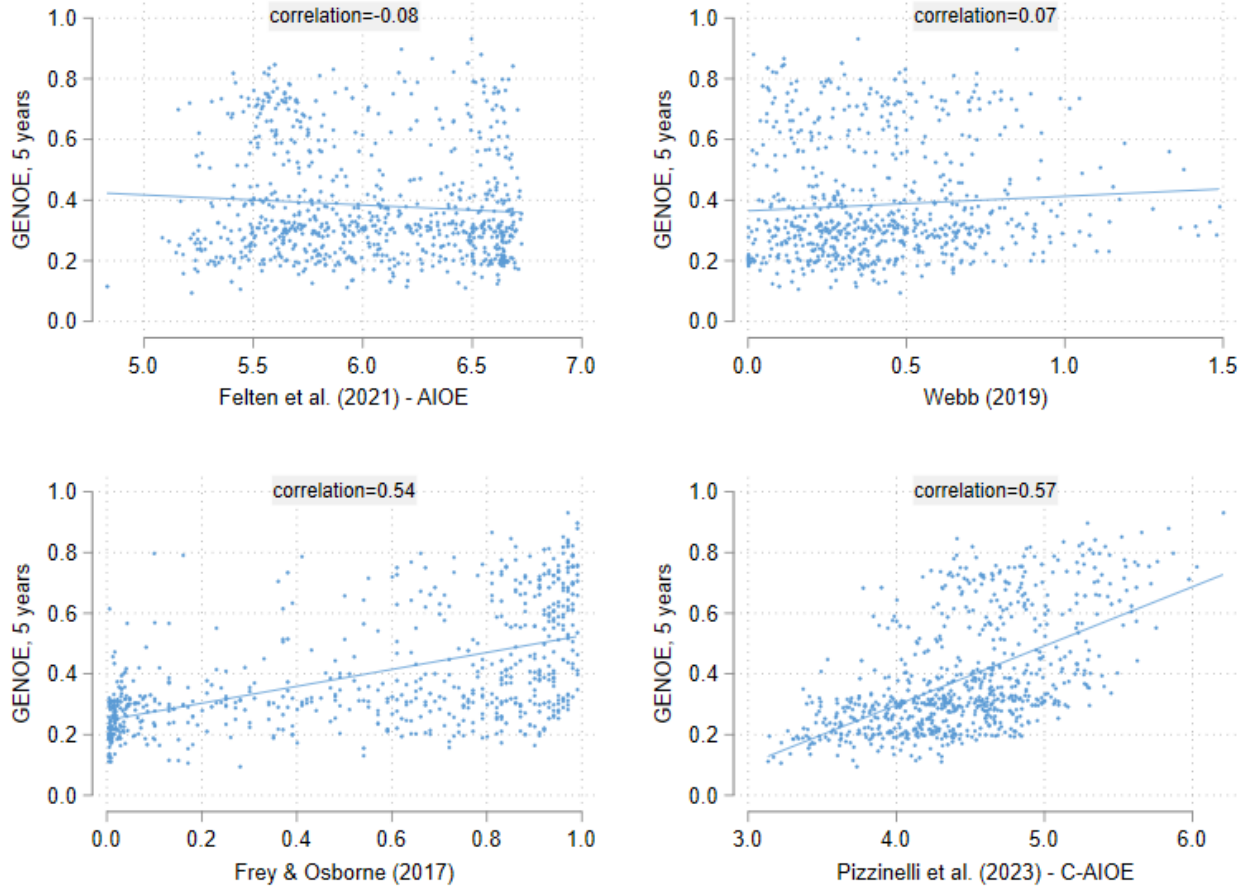
the indexes developed by Felten *et al.* (2021) and Webb (2019). However, GENOE demonstrates moderate correlations with the indexes created by Frey and Osborne (2017) and Pizzinelli *et al.* (2023). The similarities and differences can be attributed to several key factors in methodological approach and focus.

Firstly, Frey and Osborne’s index places a higher emphasis on routinization, which aligns with our follow-up surveys, indicating a similar rationale behind our LLM models’ assessment. As shown in Appendix Figure A.2, the GENOE index is higher for occupations intensive in routine manual and routine cognitive tasks. Both Pizzinelli *et al.*’s C-AIOE and Frey and Osborne’s indexes also rank high for routine-intensive occupations, with Frey and Osborne’s index higher for manual tasks and Pizzinelli *et al.*’s index higher for cognitive tasks.

Additionally, both Frey and Osborne (2017) and Pizzinelli *et al.* (2023) incorporate contextual elements beyond merely overlapping tasks or abilities with AI advancements. Pizzinelli *et al.*’s core adjustment to Felten *et al.*’s index involves considering potential complementarities between technology and labor arising from the physical and social factors of occupations. These include ethical considerations, training, and technology integration into the workplace. Similarly, Frey and Osborne’s index adopts a holistic approach akin to ours. They evaluate not just isolated tasks or abilities but the overall occupation within a specific context and task composition to determine the extent of computerization.

The divergence from Felten *et al.*’s and Webb’s indexes may be attributed to their more narrow focus on specific technological advancements or task-technology matches, without fully accounting for the broader occupational context or potential complementarities between AI and human labor.

Figure 2: Correlation of GENOE and Other Indexes of Occupational Exposure



Source: Authors' formulation based on data from O*NET Felten *et al.* (2021); Webb (2019); Pizzinelli *et al.* (2023). Note: This figure displays the GENOE for the next one, five, and 10 years. The index measures the likelihood that occupations will be displaced by AI within each time horizon.

3.3 Alternative Methodologies

We explore how GENOE estimates vary under different methodologies and prompt choices. Our baseline specification involves providing the model with the occupation name and a vector of tasks and asking for an estimate of the likelihood of job replacement. We then introduce six alternative indexes, each employing a distinct approach:

Blind Index: This approach removes the occupation name, relying solely on the task vector for estimations. This allows us to isolate the impact of task-specific factors from occupation-specific context factors in the model's predictions and analyze if the absence of the occupation name significantly alters the estimates.

Task-by-Task Index: Instead of aggregating tasks under an occupation, we estimate the likeli-

hood of replacement for each task individually. These estimates are then averaged and weighted according to the relevance and importance of each task within the occupation, providing a granular view of task vulnerability.

Ethical Emphasis Index: Here, we explicitly ask the model to consider the ethical repercussions and potential social resistance to replacing certain tasks with AI. This approach highlights whether explicitly including ethical considerations causes a significant deviation from the baseline, which may already implicitly factor in such considerations.

Ethical Exclusion Index: This index explicitly asks the model not to consider ethical, regulatory, or social impediments to replacing tasks with AI. The objective of this index, compared to the previous one, is to measure the increase in the index when these considerations are excluded.

LLM-Specific Index: This variant focuses specifically on large language model (LLM) technologies. By narrowing the scope to LLM innovations, we assess the unique impact of advancements in natural language processing and related applications on occupational exposure, contrasting it with the general AI exposure measured in the baseline.

Non-AI Technologies Index: We estimate the probability of task replacement by non-AI technologies such as digital tools, robotics, and computer software. This provides a comparative baseline, distinguishing the impact of traditional technologies from AI-driven automation.

AI-Only Index: This index is derived by subtracting the Non-AI Technologies Index from the baseline GENOE, capturing the incremental exposure attributable solely to AI advancements. It allows us to identify occupations where AI-specific risks have significantly increased.

Table 4 provides the descriptive statistics of these alternative indexes, their correlation with the baseline GENOE, and other existing exposure indexes, and the extent to which they differ from the baseline. Both the Blind Index and the Ethical Emphasis Index do not deviate significantly from the baseline, although the mean differences are statistically significant. This indicates that there are slight changes in the reasoning of the model when the occupation name is omitted or when ethical considerations are explicitly prompted, but most of these factors are already incorporated in the baseline index. The overall correlations of these indexes with the baseline GENOE are 0.95 and 0.99, respectively. This consistency is supported by follow-up surveys, where ethical considerations and occupation-specific criteria were crucial elements for many of the model's assessments.

GENOE's ability to reflect ethical, regulatory, and social considerations in its assessment of AI's potential to replace human labor, even when not explicitly directed to do so, is a critical aspect of our argument. This aspect was initially confirmed by our follow-up survey analysis and is also supported by the comparisons with alternative indexes. Notably, when we explicitly instructed the model to include these factors, the resulting index showed only a minor reduction of one percentage point over the 5-year horizon. However, when we asked the model to exclude these considerations, the index increased by four percentage points, from 0.38 to 0.42. A similar pattern is observed for the 10-year horizon, as illustrated in Appendix Figure A.3, though the effect is less

marked over the one-year horizon.

For the Task-by-Task Index, the average exposure increases substantially to 0.56, compared to 0.38 in the baseline. This highlights the fact that exposure to individual tasks does not linearly translate into the same exposure to job replacement. In other words, some occupations have high task-exposure but are embedded in occupation-specific contexts that mitigate overall exposure. Additionally, as previously mentioned, the disaggregated approach does not account for how the interrelation of tasks may result in different levels of exposure to replacement. However, it is worth noting that this index, despite having higher values, exhibits a high correlation with the baseline index (0.89).

The specific LLM Index estimations suggest that although a significant source of exposure emerges from the enhanced capabilities of language model technologies (with an index value of 0.20), it accounts for only 54% of the general AI index. Thus, the model considers broader potential applications of AI in the baseline scenario. The overall correlation with the GENOE is modest, at about 0.47. It is also noteworthy that the LLM Index is more correlated with some of the existing indexes of AI exposure, such as Felten et al. (2021)'s AIOE and Pizzinelli et al. (2023)'s C-AIOE. This may reflect the fact that these indexes also capture the specific impacts of language model technologies and their applications.

Exposure to non-AI technologies is also considerable, with an average index value of about 0.34. Interestingly, most occupations exposed to AI are also exposed to other types of technologies, as suggested by the high correlation of the Non-AI Index with the baseline (0.95). Additionally, this index is most correlated with the Frey and Osborne (2017) measure of exposure, probably because that index relates more closely to the broader exposure to computerization, including non-AI innovations.

The final "Only AI" Index allows us to determine which occupations will experience a higher increase in exposure to technology specifically due to AI. The average value of this index is low, at 0.04, but it reaches up to 0.39 for the highest values. Occupations with the highest values include clerical workers and data-intensive and writing-intensive roles such as translators, credit analysts, insurance underwriters, customer service representatives, paralegals, loan officers, and proofreaders.